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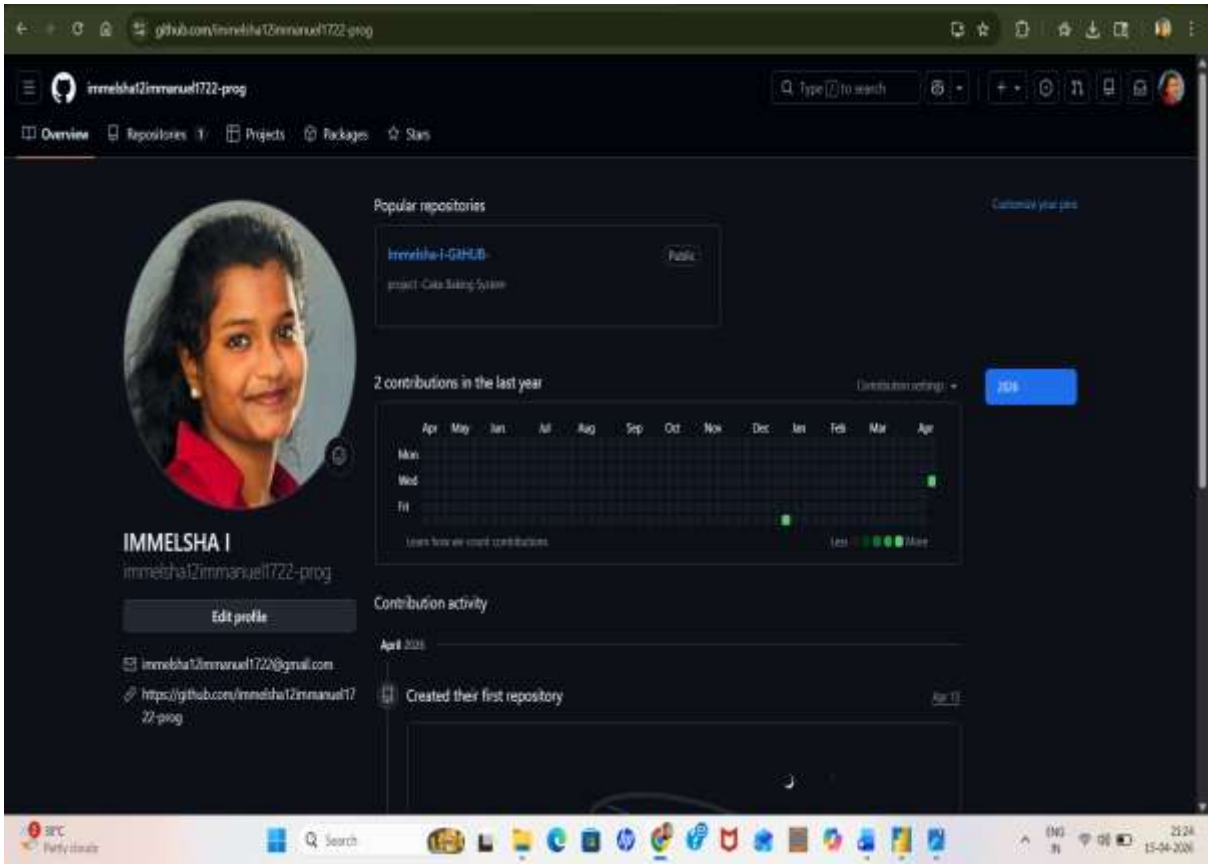
VTP NUMBER: VTP4938

GITHUBLINK : https://github.com/immelsha12immanuel1722-prog



//Build simple java application with class object and methods and make use of package creation

PROGRAM:

Cake.java

```
package startupcake;

import java.util.*;
import java.io.*;

public class Cake {

String cakeName;
int quantity;

void getCakeDetails() {
Scanner sc = new Scanner(System.in);

System.out.print("Enter Cake Name: ");
cakeName = sc.nextLine();

System.out.print("Enter Quantity: ");
quantity = sc.nextInt();
}

void bakeCake() {
System.out.println("Baking " + quantity + " " + cakeName + " cake(s)...");
}
}
```

CakeApp.java

```
package startupcake;
import java.util.*;
import java.io.*;

public class CakeApp {

public static void main(String[] args) {

Cake c = new Cake();   // object creation
c.getCakeDetails();
c.bakeCake();

System.out.println("Startup Cake Baking System Finished!");
}
```

}

Output:

```
Command Prompt - cmd

Microsoft Windows [Version 10.0.26200.8117]
(c) Microsoft Corporation. All rights reserved.

C:\Users\immel>cmd
Microsoft Windows [Version 10.0.26200.8117]
(c) Microsoft Corporation. All rights reserved.

C:\Users\immel>cd C:\JavaPrograms

C:\JavaPrograms>javac startupcake\Cake.java startupcake\CakeApp.java

C:\JavaPrograms>java startupcake.CakeApp
Enter Cake Name: Pista
Enter Quantity: 1
Baking 1 Pista cake(s)...
Startup Cake Baking System Finished!

C:\JavaPrograms>
```

//Inheritance and Interface

PROGRAM

Bakeable.java

```
package startupcake;

interface Bakeable {
   void bake();
}
```

BakeryItem.java

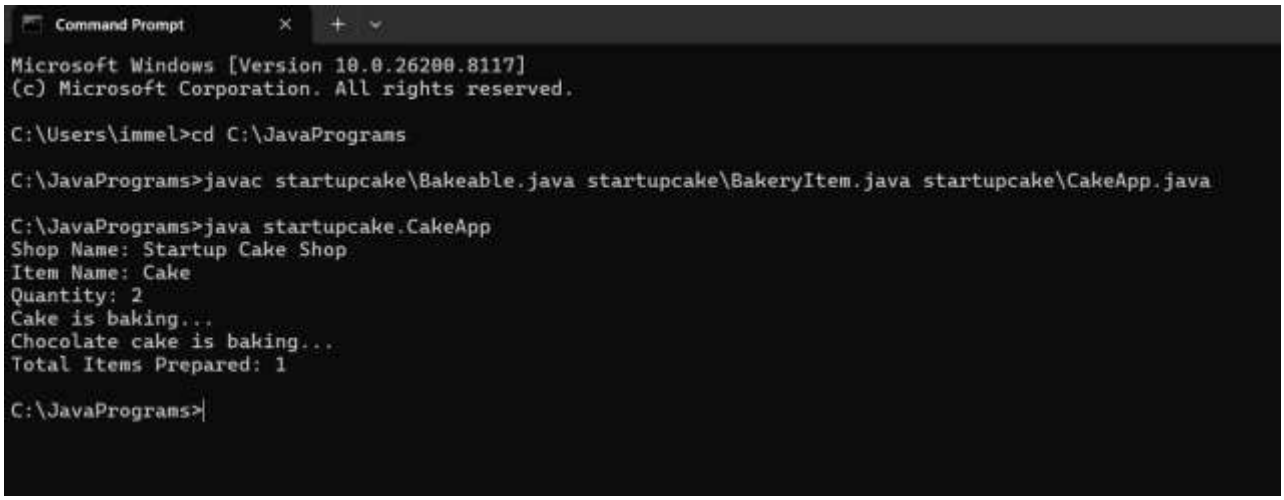
```
package startupcake;

class BakeryItem {

   final String shopName = "Startup Cake Shop";   // final keyword
   static int itemCount = 0;                      // static keyword

   String name;
   int quantity;

   // Constructor
   BakeryItem(String name, int quantity) {
      this.name = name;       // this keyword
      this.quantity = quantity;
      itemCount++;
   }

   void display() {
      System.out.println("Shop Name: " + shopName);
      System.out.println("Item Name: " + name);
      System.out.println("Quantity: " + quantity);
   }
}
```

CakeApp.java

```
package startupcake;

class Cake extends BakeryItem implements Bakeable {

   // Constructor using super
   Cake(String name, int quantity) {
      super(name, quantity);   // super keyword
   }
```

```
    // Method Overriding
    public void bake() {
        System.out.println("Cake is baking...");
    }

    // Method Overloading
    void bake(String flavor) {
        System.out.println(flavor + " cake is baking...");
    }
}

public class CakeApp {

    public static void main(String[] args) {

        Cake c1 = new Cake("Cake", 2);

        c1.display();       // inherited method
        c1.bake();          // overridden method
        c1.bake("Chocolate");// overloaded method

        System.out.println("Total Items Prepared: " + BakeryItem.itemCount);
    }
}
```

Output :



//Exception Handling Mechanism in java

PROGRAM

OnlineShop.java

```
package onlineshop;

import java.util.*;

// User-defined Exception
class InvalidOrderException extends Exception {
   InvalidOrderException(String msg) {
      super(msg);
   }
}

public class OnlineShop {

   public static void main(String[] args) {

      Scanner sc = new Scanner(System.in);

      try {
         System.out.print("Enter product price: ");
         int price = sc.nextInt();

         System.out.print("Enter quantity: ");
         int qty = sc.nextInt();

         // User-defined exception
         if (qty <= 0) {
            throw new InvalidOrderException("Quantity must be greater than 0");
         }

         int total = price * qty;

         System.out.println("Total Bill: " + total);

         // Pre-defined exception (ArithmeticException)
         int discount = total / 0;   // Force error

         System.out.println("Discount: " + discount);
      }

      catch (InvalidOrderException e) {
         System.out.println("Order Error: " + e.getMessage());
      }
```

```
        catch (ArithmeticException e) {
            System.out.println("Math Error: Cannot divide by zero");
        }

        catch (Exception e) {
            System.out.println("General Error: " + e);
        }

        finally {
            System.out.println("Thank you for shopping!");
        }
    }
}
```

Output:

```
Command Prompt

Microsoft Windows [Version 10.0.26200.8117]
(c) Microsoft Corporation. All rights reserved.

C:\Users\immel>cd C:\JavaPrograms

C:\JavaPrograms>javac onlineshop\OnlineShop.java

C:\JavaPrograms>java onlineshop.OnlineShop
Enter product price: 465
Enter quantity: 2
Total Bill: 930
Math Error: Cannot divide by zero
Thank you for shopping!

C:\JavaPrograms>
```

//String and java collections

Program code
CakeCollection.java

```
package cakecollection;

import java.util.*;

public class CakeCollection {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // List
        List<String> cakeList = new ArrayList<>();

        // Set (no duplicates)
        Set<String> cakeSet = new HashSet<>();

        // Queue (orders)
        Queue<String> orderQueue = new LinkedList<>();

        // Map (id → cake)
        Map<Integer, String> cakeMap = new HashMap<>();

        System.out.print("Enter number of items to order: ");
        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 1; i <= n; i++) {

            System.out.print("Enter cake name: ");
            String cake = sc.nextLine();

            // String methods
            cake = cake.trim();                // remove spaces
            cake = cake.toUpperCase();         // uppercase

            // Add to collections
            cakeList.add(cake);
            cakeSet.add(cake);
            orderQueue.add(cake);
            cakeMap.put(i, cake);

            System.out.println(cake + " cake added to queue");
        }
```

```
        // Display results
        System.out.println("\n--- ORDER DETAILS ---");

        System.out.println("List (All Items): " + cakeList);
        System.out.println("Set (Unique Items): " + cakeSet);

        System.out.println("Queue (Processing Order): " + orderQueue);
        System.out.println("Serving: " + orderQueue.poll());

        System.out.println("Map (ID -> Cake): " + cakeMap);

        sc.close();
    }
}
```

Output:

```
C:\JavaPrograms>cd C:\JavaPrograms

C:\JavaPrograms>javac cakecollection\CakeCollection.java

C:\JavaPrograms>java cakecollection.CakeCollection
Enter number of items to order: 2
Enter cake name: vanilla cake
VANILLA CAKE cake added to queue
Enter cake name: choco truffle
CHOCO TRUFFLE cake added to queue

--- ORDER DETAILS ---
List (All Items): [VANILLA CAKE, CHOCO TRUFFLE]
Set (Unique Items): [VANILLA CAKE, CHOCO TRUFFLE]
Queue (Processing Order): [VANILLA CAKE, CHOCO TRUFFLE]
Serving: VANILLA CAKE
Map (ID -> Cake): {1=VANILLA CAKE, 2=CHOCO TRUFFLE}

C:\JavaPrograms>
```

//Threading

PROGRAM:

CakeThreadDemo.java

```
package threaddemo;

// Shared resource (Kitchen)
class Bakery {

   // synchronized → one order at a time
   synchronized void bakeCake(String customer, String cake) {

      System.out.println("Now baking for " + customer);

      try {
         Thread.sleep(1000);
      } catch (Exception e) {}

      System.out.println("Preparing " + cake + " cake");

      try {
         Thread.sleep(1000);
      } catch (Exception e) {}

      System.out.println(customer + " served\n");
   }
}

// Thread class
class CustomerThread extends Thread {

   Bakery bakery;
   String customerName;
   String cakeName;

   CustomerThread(Bakery bakery, String customerName, String cakeName) {
      this.bakery = bakery;
      this.customerName = customerName;
      this.cakeName = cakeName;
   }

   public void run() {
      bakery.bakeCake(customerName, cakeName);
   }
}

// Main class
public class CakeThreadDemo {
```

```
    public static void main(String[] args) {

        Bakery b = new Bakery();   // shared resource

        // Multiple customers (threads)
        CustomerThread t1 = new CustomerThread(b, "Selva Lakshmi", "Chocolate");
        CustomerThread t2 = new CustomerThread(b, "Akash", "Plum");
        CustomerThread t3 = new CustomerThread(b, "Vani", "Vanilla");

        t1.start();
        t2.start();
        t3.start();
    }
}
```

OUTPUT:

```
C:\JavaPrograms>cd C:\JavaPrograms

C:\JavaPrograms>javac threaddemo\CakeThreadDemo.java

C:\JavaPrograms>java threaddemo.CakeThreadDemo
Now baking for Selva Lakshmi
Preparing Chocolate cake
Selva Lakshmi served

Now baking for Akash
Preparing Plum cake
Akash served

Now baking for Vani
Preparing Vanilla cake
Vani served


C:\JavaPrograms>
```

//Networking

Program:

```
package networkdemo;

import java.net.*;
import java.io.*;

// MAIN CLASS
public class OneNetworkDemo {

   // -------- TCP SERVER THREAD --------
   static class TCPServer extends Thread {
      public void run() {
         try {
            ServerSocket ss = new ServerSocket(5000);
            Socket s = ss.accept();

            DataInputStream din = new DataInputStream(s.getInputStream());
            DataOutputStream dout = new DataOutputStream(s.getOutputStream());

            String order = din.readUTF();
            System.out.println("TCP Server: Order Received -> " + order);

            dout.writeUTF("Order Received (TCP)");

            s.close();
            ss.close();
         } catch (Exception e) {
            System.out.println(e);
         }
      }
   }

   // -------- TCP CLIENT THREAD --------
   static class TCPClient extends Thread {
      String order;

      TCPClient(String order) {
         this.order = order;
      }

      public void run() {
         try {
            Thread.sleep(500); // wait for server

            Socket s = new Socket("localhost", 5000);
```

```
            DataOutputStream dout = new DataOutputStream(s.getOutputStream());
            DataInputStream din = new DataInputStream(s.getInputStream());

            dout.writeUTF(order);

            String reply = din.readUTF();
            System.out.println("TCP Client: " + reply);

            s.close();
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}

// -------- UDP RECEIVER THREAD --------
static class UDPReceiver extends Thread {
    public void run() {
        try {
            DatagramSocket ds = new DatagramSocket(6000);

            byte[] buf = new byte[1024];
            DatagramPacket dp = new DatagramPacket(buf, buf.length);

            ds.receive(dp);

            String msg = new String(dp.getData(), 0, dp.getLength());
            System.out.println("UDP Receiver: Order Received -> " + msg);

            ds.close();
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}

// -------- UDP SENDER THREAD --------
static class UDPSender extends Thread {
    String order;

    UDPSender(String order) {
        this.order = order;
    }

    public void run() {
        try {
            Thread.sleep(500); // wait for receiver
```

```
            DatagramSocket ds = new DatagramSocket();

            byte[] buf = order.getBytes();
            InetAddress ip = InetAddress.getByName("localhost");

            DatagramPacket dp =
                  new DatagramPacket(buf, buf.length, ip, 6000);

            ds.send(dp);

            System.out.println("UDP Sender: Order Sent");

            ds.close();
         } catch (Exception e) {
            System.out.println(e);
         }
      }
   }

   // -------- MAIN METHOD --------
   public static void main(String[] args) throws Exception {

      BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

      System.out.print("Enter Order: ");
      String order = br.readLine();

      // TCP
      TCPServer ts = new TCPServer();
      TCPClient tc = new TCPClient(order);

      // UDP
      UDPReceiver ur = new UDPReceiver();
      UDPSender us = new UDPSender(order);

      // Start all
      ts.start();
      tc.start();
      ur.start();
      us.start();
   }
}
```

OUTPUT:

```
C:\JavaPrograms>cd C:\JavaPrograms

C:\JavaPrograms>javac networkdemo\OneNetworkDemo.java

C:\JavaPrograms>java networkdemo.OneNetworkDemo
Enter Order: Plum cake
UDP Sender: Order Sent
UDP Receiver: Order Received -> Plum cake
TCP Server: Order Received -> Plum cake
TCP Client: Order Received (TCP)

C:\JavaPrograms>
```

//AWT AND JavaFX Components

PROGRAM

```
import java.awt.*;
import java.awt.event.*;

public class CakeOrderAWT extends Frame implements ActionListener {

Image bgImage;

Label nameLabel, cakeLabel, quantityLabel, dateLabel, phoneLabel, paymentLabel;

TextField nameText, quantityText, dateText, phoneText;

Choice cakeChoice, paymentChoice;

Button addButton, removeButton, clearButton;

TextArea orderArea;

GridBagConstraints gbc;

CakeOrderAWT(){

bgImage = Toolkit.getDefaultToolkit().getImage("cake.jpg");

setTitle("Cake Baking Order System");
setSize(500,450);

setLayout(new GridBagLayout());

gbc = new GridBagConstraints();
gbc.insets = new Insets(10,10,10,10);

// Labels
nameLabel = new Label("Customer Name:");
cakeLabel = new Label("Cake Order:");
quantityLabel = new Label("Quantity:");
dateLabel = new Label("Buying Date:");
phoneLabel = new Label("Phone Number:");
paymentLabel = new Label("Payment:");

// TextFields
nameText = new TextField(15);
quantityText = new TextField(5);
dateText = new TextField(10);
```

```
phoneText = new TextField(12);

// Cake options
cakeChoice = new Choice();
cakeChoice.add("Chocolate");
cakeChoice.add("Black Forest");
cakeChoice.add("Red Velvet");
cakeChoice.add("Vanilla");

// Payment options
paymentChoice = new Choice();
paymentChoice.add("Credit Card");
paymentChoice.add("Debit Card");
paymentChoice.add("UPI");
paymentChoice.add("Cash");

// Buttons
addButton = new Button("Add Order");
removeButton = new Button("Remove Orders");
clearButton = new Button("Clear");

// Order area
orderArea = new TextArea(5,35);

// Row 1
gbc.gridx=0; gbc.gridy=0;
add(nameLabel,gbc);

gbc.gridx=1;
add(nameText,gbc);

// Row 2
gbc.gridx=0; gbc.gridy=1;
add(cakeLabel,gbc);

gbc.gridx=1;
add(cakeChoice,gbc);

// Row 3
gbc.gridx=0; gbc.gridy=2;
add(quantityLabel,gbc);

gbc.gridx=1;
add(quantityText,gbc);

// Row 4
gbc.gridx=0; gbc.gridy=3;
```

```
add(dateLabel,gbc);

gbc.gridx=1;
add(dateText,gbc);

// Row 5
gbc.gridx=0; gbc.gridy=4;
add(phoneLabel,gbc);

gbc.gridx=1;
add(phoneText,gbc);

// Row 6
gbc.gridx=0; gbc.gridy=5;
add(paymentLabel,gbc);

gbc.gridx=1;
add(paymentChoice,gbc);

// Buttons
gbc.gridx=0; gbc.gridy=6;
add(addButton,gbc);

gbc.gridx=1;
add(removeButton,gbc);

gbc.gridx=2;
add(clearButton,gbc);

// Order display
gbc.gridx=0; gbc.gridy=7;
gbc.gridwidth=3;
add(orderArea,gbc);

// Events
addButton.addActionListener(this);
removeButton.addActionListener(this);
clearButton.addActionListener(this);

setVisible(true);
}

// Background image
public void paint(Graphics g){
g.drawImage(bgImage,0,0,getWidth(),getHeight(),this);
}
```

```
// Button actions
public void actionPerformed(ActionEvent e){

if(e.getSource()==addButton){

String order =
nameText.getText()+" - "+
cakeChoice.getSelectedItem()+" - Qty:"+quantityText.getText()+" - "+
"Date:"+dateText.getText()+" - "+
phoneText.getText()+" - "+
paymentChoice.getSelectedItem();

orderArea.append(order+"\n");

}

if(e.getSource()==removeButton){
orderArea.setText("");
}

if(e.getSource()==clearButton){
nameText.setText("");
quantityText.setText("");
dateText.setText("");
phoneText.setText("");
}

}

public static void main(String args[]){
new CakeOrderAWT();
}

}
```

TERMINAL ··· java

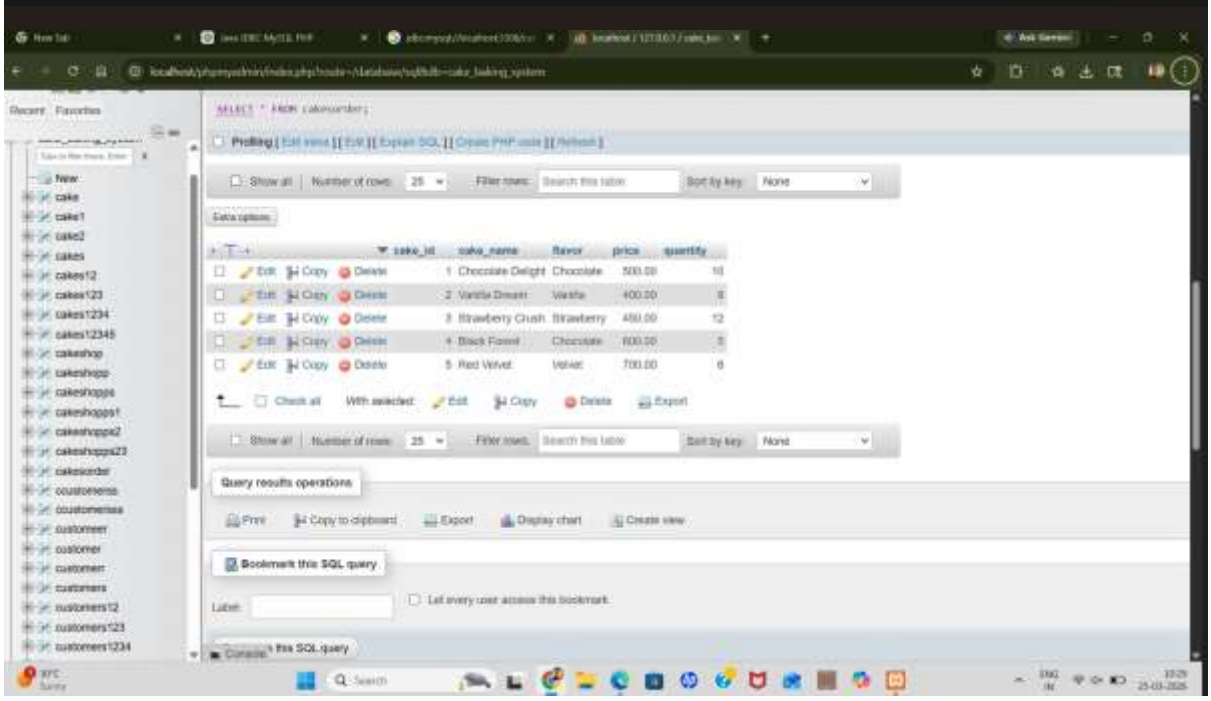
```
PS C:\Users\immel\OneDrive\Documents\Desktop\CakeOrderSystem> javac CakeOrderAWT.java
PS C:\Users\immel\OneDrive\Documents\Desktop\CakeOrderSystem> java CakeOrderAWT
```

//To develop a cakebaking system using java as frondend and mysql as backend

PROGRAM:

```
CREATE TABLE cakes (
   cake_id INT PRIMARY KEY AUTO_INCREMENT,
   cake_name VARCHAR(50),
   quantity INT,
   price DOUBLE,
   order_date DATE
);

INSERT INTO cakes (cake_name, quantity, price, order_date) VALUES
('Chocolate Cake', 2, 500, '2026-03-25'),
('Vanilla Cake', 1, 300, '2026-03-26'),
('Black Forest Cake', 3, 700, '2026-04-01'),

INSERT INTO customers (customer_name, phone, address, order_date) VALUES
('Arun Kumar', '9876543210', 'Salem', '2026-03-25'),
('Priya Sharma', '9123456780', 'Chennai', '2026-03-26'),
// Program Code:

import java.sql.Connection;
import java.sql.DriverManager;

public class DBConnection {

   public static Connection getConnection() {
      Connection con = null;

      try {
         Class.forName("com.mysql.cj.jdbc.Driver");

         con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/cake_baking_system",
            "root",
            ""
         );

         System.out.println("Connection to database");
      }
      catch (Exception e) {
         System.out.println("Error: " + e);
```

```
        }
        return con;
    }
}
Public class MainApp {
    public static void main(String[] args) {

        try {
            Connection con = DBConnector.getConnection();

            System.out.println("----- CAKES -----");

            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("select * from cakeorders");

            while (rs.next()) {
                System.out.println(
                    rs.getInt("cake_id") + " | " +
                    rs.getString("cake_name") + " | " +
                    rs.getString("flavor") + " | " +
                    rs.getDouble("price") + " | " +
                    rs.getInt("Quantity")
                );
            }

            System.out.println("\n----- CUSTOMER -----");

            ResultSet rs2 = st.executeQuery("Select * from customers detail");

            while (rs2.next()) {
                System.out.println(
                    rs2.getInt("customer_ID") + " | "
                );
            }

        }
    }
}
rs2.getString("phone") + " | " +
rs2.getString("address"));

}

} catch (Exception e) {
    System.out.println(e);
}
```

OUTPUT:

```
java.lang.ClassNotFoundException: com.mysql.cj.jdbc.Driver
java.lang.ClassNotFoundException: com.mysql.cj.jdbc.Driver
PS C:\CakeSystem> java -cp ".;mysql-connector-j-9.6.0.jar" CakeSystem
>>
java.lang.ClassNotFoundException: com.mysql.cj.jdbc.Driver
PS C:\CakeSystem>
```